

Vincent Mastain

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Education

PhD Thesis - Scalable Translation Validation for High Performance Computing and Machine Learning At INRIA - Ecole Normale Supérieure	Lyon, France 2025 - 2028
Engineering Degree in Computer Science - Artificial Intelligence At CY-Tech software engineering school	Cergy, France 2022 - 2024
Integrated Preparatory Class At CY-Tech software engineering school	Cergy, France 2019 - 2022
Scientific Baccalaureate - Mathematics Option - Highest Honors Lycée Voltaire	Wingles, France 2019

Skills

- **Programmation :** Java | Python | C++ | C# | Javascript | Dart | OCaml | C
- **Technologies and tools :** OOP | Design Patterns | Testing | Unix | JUnit5 | Git/GitHub | Software Engineering | Docker | Kubernetes | Keras | PyTorch | Deep Learning | NLP | Flutter | ReactJS | ExpressJS | Angular | MySQL

Experience

 PhD Thesis - Scalable Translation Validation for High Performance Computing and Machine Learning	Lyon, France September 2025 - August 2028
• Conducted a literature review on parallel program equivalence and parallel program analysis techniques. • Defined the happens-before relation for OpenMP programs and developed the corresponding algorithm. • Implemented this algorithm using PoCo, the Polyhedral Compiler developed by my PhD supervisor Christophe Alias.	
 Internship - Quality Assurance & Automation Engineer	Levallois-Perret, France May 2024 - October 2024
• Maintenance and addition of automated tests for various projects using an internal Selenium-based framework. • Creation of a pipeline enabling the automation of test environment creation and test execution. • Ticket writing and proactive communication with developers to follow up on bug reports. • Experimentation with the Behavior Driven Development (BDD) methodology to assess its impact on improving the quality of automated tests. • Analysis of BDD prototyping results to optimize testing processes and team collaboration.	
 Research Student - Development of a Testing Framework for Videogames	Montréal, Canada May 2023 - August 2023

As a research student at ÉTS (École de Technologie Supérieure), I worked under the supervision of Fabio Petrillo, actively engaging in the exciting field of video game testing using reinforcement learning techniques. My main responsibility was to design and implement a comprehensive framework for developers to evaluate and test their games.

- Designed and developed a testing software in Python, integrating video games and based on the Behavior Driven Development (BDD) method, similar to the Cucumber tool.
- Developed and implemented reinforcement learning algorithms.
- Created unit tests to ensure the software's full functionality.
- Developed video games to test the software in real-world scenarios.
- Conducted a literature review on reinforcement learning and video game testing.

Scientific Publications

<u>A Behavior-driven Development and Reinforcement Learning Approach for Videogame Automated Testing</u>	Lisbonne, Portugal, ICSE - GAS, April 15, 2024
Paper proposing a solution based on the BDD methodology to improve the quality of video games at launch. Presented at the GAS workshop - ICSE 2024. We are currently working on generalizing this approach using the GODOT game engine. (Case studies on github)	

Languages

English Professional working proficiency - TOEIC 965 / 990 **French** Native proficiency